Armon × Gauthier × Rick × Julien

1 Literature

- Sahuc et al. (2024): Integrates climate into a New Keynesian (NK) framework. Disentangles *greenflation* from *climateflation* and analyzes the optimal monetary response to climate shocks.
- Bergin and Corsetti (2023): Demonstrates that tariff shocks differ from traditional supply shocks. Using a stylized supply chain dimension, they show that monetary expansion is the optimal response to offset the distortionary nature of tariffs.
- Bergin and Corsetti (2025a): Analyzes a two-sector model (differentiated/sticky prices vs. non-differentiated/flexible prices). Highlights that optimal monetary policy requires coordination, as price reactions vary significantly across sectors.
- Bergin and Corsetti (2025b): Explores how monetary policy favors reallocation induced by tariffs. Expansionary policy triggers depreciation to offset relative price increases. Emphasizes the role of comparative advantage and currency dominance in a tariff war scenario.
- Ambec et al. (2024): Examines welfare effects and efficiency of CBAM in an industry producing homogeneous polluting goods. Shows that export rebates and marginal cost heterogeneity determine whether a no-trade equilibrium emerges.
- Clausing et al. (2025): Quantifies CBAM effects (focusing on steel/aluminum) using trade and production elasticities. CBAM shifts the marginal abatement cost (MAC) and provides incentives for foreign regulatory adaptation.
- Coster et al. (2024): Develops an input sourcing model to estimate carbon leakage. Includes a disutility term for climate damages. Highlights the impact of CBAM on firm-level sourcing decisions under monopolistic competition.

2 General questions

- CBAM applies only to specific products; how does the nature of these products (intermediates/final, differentiated/non differentiated) change the transmission?
- Does CBAM trigger technological progress? What effect on abatement decisions?
- Should we/How explicitly model the reorganization of supply chains triggered by carbon pricing?
- Bergin & Corsetti emphasize cooperative Ramsey policies. Does it apply to CBAM? Interesting as a point of comparison but I can imagine retaliation or unilateral policy to offset CBAM.

- Monetary policy tries to offset tariff distorsions through exchange rates. CBAM actually want to decrease trade of carbon intensive products, how this changes the optimal policy?
- How are CBAM revenues used?
- How CBAM path affects price-setting by firms?
- Do we need a damage function? Country-specific damages? Shouldn't the damage affect both final goods and intermediate goods production?

3 Two country NKC model: everything tradable, flexible price final good

3.1 Final Good Sector

Two countries, home (h) and foreign (f). Production uses domestic and imported intermediates:

$$y_t(x) = \alpha_x [G_t(x)]^{\zeta} l_t^{1-\zeta}$$

where $G_t(x)$ is a CES aggregator with substitutability η:

$$G_t(x) = \left(\int_0^{l_t(h)} y_{j,t}(h)^{\frac{\eta-1}{\eta}} dj + \int_0^{l_t(f)} y_{j,t}(f)^{\frac{\eta-1}{\eta}} dj \right)^{\frac{\eta}{\eta-1}}$$

The profit of the final good firm is, taking into account iceberg cost of trade (τ_i):

$$\begin{aligned}
\Pi_t(h) = p_t(h)\alpha_h l_t^{1-\zeta} \bigg[G_t(h) \bigg]^{\zeta} &- \int_0^{l_t(h)} p_{j,t}(h) y_{j,t}(h) dj \\
&- e_t \int_0^{l_t(f)} (1+\tau_i) p_{j,t}(f) y_{j,t}(f) dj \\
&- \int_0^{l_t(f)} \underbrace{\max \left\{ \underbrace{(1+\tau_{e,t}(h))}_{\text{carbon tax home}} - \underbrace{(1+\tau_{e,t}(f))}_{\text{carbon tax paid}}, 0 \right\}}_{\text{CBAM}} \underbrace{\sigma_t(f)(1-\mu_{j,t}(f)) y_{j,t}(f)}_{\text{emission for production } y_{j,t}} dj
\end{aligned}$$

CBAM enters profit through a carbon tax on top of foreign one, meaning on $p_{i,j,t}(f)$.

Question:

- Should these prices be sticky?
- Do we need a CES production function?

3.2 Intermediate Good Sector

Intermediate goods are produced by monopolistic firms:

$$y_{j,t}(x) = \Gamma_t(x) l_{j,t}(x)^{1-\alpha} (n_{j,t}(x)^d)^\alpha$$

where the TFP term $\Gamma_t(x) = z_t(x)\Phi_x(m_t)$ includes the damage function $\Phi_x(m_t) = \exp(-\gamma_x(m_t - m_{1750}))$. Emissions follow:

$$e_{j,t}(x) = \sigma_t(x)(1 - \mu_{j,t}(x)) y_{j,t}(x)$$

The country-specific abatement cost is:

$$C^a_{j,t}(x) = \theta_{1,t}(x)\mu_{j,t}^{\theta_2(x)} y_{j,t}(x) \quad \text{where} \quad \theta_{1,t}(x) = \frac{p_b(x)}{\theta_2(x)}(1 - \delta_{pb}(x))^{t-t_0}\sigma_t(x)$$

With a country specific emission level, carbon taxation and abatement costs it is possible how exports and imports are affected by different levels of carbon taxation with and without carbon adjustment mechanism.
We can calibrate emissions and costs to match worldwide differences in marginal costs and emission-intensity in production processes.

Profit of intermediates firms is such :

$$\max \pi_{j,t}(x) = mc_{j,t}(x) y_{j,t}(x) - w_t(x)\left(\frac{y_{j,t}(x)}{\Gamma_t(x)}\right)^{1/\alpha} - \underbrace{C^a_{j,t}(x)}_{\text{abatement}} - \underbrace{\tau_{e,t}(x)\sigma_t(x)(1 - \mu_{j,t}(x)) y_{j,t}(x)}_{\text{carbon tax}}$$

Firms face Rotemberg adjustment costs:

$$C^p_{j,t}(x) = \frac{\kappa}{2}\left(\frac{p_{j,t}(x)}{p_{j,t-1}(x)} - \pi^\star\right)^2 \frac{y_t(x)}{l_t(x)}$$

Death shock (exogenous exit of firm, Bilbiie et al. (2012); Del Negro et al. (2023)) with probability $\vartheta \in (0,1)$. $\omega_{j,t} = \{0,1\}$ indicates if the firm exited the market or not. Intermediate goods firms producer maximize:

$$max_{\{p_{j,t}\}} \mathbb{E}_t \left\{ \sum_{s=0}^{\infty} \beta^s \omega_{j,t+s} \left(y_{j,t+s}\frac{p_{j,t+s}}{p_{t+s}} - \varepsilon_{p,t+s} mc_{t+s} y_{j,t+s} - C^p_{j,t+s} \right) \right\}$$

Question: Free entry in both countries + exogenous death shocks. How to manage firm entry?

3.3 Households

Households maximize lifetime utility subject to employment status $\theta \in \{0,1\}$:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{c_{i,t}^{1-\sigma}}{1-\sigma} - \psi_t \frac{n_{i,t}^{1+\sigma_n}}{1+\sigma_n} \right\}$$

The consumption aggregate:

$$c_{i,t} = \left(c_{i,t}(h)^{\frac{\phi-1}{\phi}} + c_{i,t}(f)^{\frac{\phi-1}{\phi}} \right)^{\frac{\phi}{\phi-1}}$$

Price index:

$$P_t = \left[p_t(h)^{1-\phi} + (e_t p_t(f))^{1-\phi} \right]^{\frac{1}{1-\phi}}$$

Budget Constraints:

$$\theta_{i,t} = 1: \quad c_{i,t} + b_{i,t}(h) + e_t b_{i,t}(f) + T^S_{i,t} = \frac{r_{t-1}(h)}{\pi_t} b_{i,t-1}(h) + e_t \frac{r_{t-1}(f)}{\pi_t} b_{i,t-1}(f) + \Pi_{i,t} + w_t n_{i,t} + \frac{T^e_{i,t}}{1-\omega}$$

$$\theta_{i,t} = 0: \quad c_{i,t} + b_{i,t}(h) + e_t b_{i,t}(f) = \frac{r_{t-1}(h)}{\pi_t} b_{i,t-1}(h) + e_t \frac{r_{t-1}(f)}{\pi_t} b_{i,t-1}(f) + d_{i,t}$$

Question: Should we add money holdings M_t/P_t?

4 Exercises

- Effect of the introduction of CBAM on inflation.
- Non cooperative monetary policy ==> can foreign country offset the CBAM through devaluation?
- Monetary policy : comparison Taylor rule, Ramsey cooperative, maximization of national welfare given policy rule in the foreign country?
- Difference between CBAM and tariff?
- No CBAM, introduction of carbon tax, observe the production reallocation towards other country and model carbon leakage.

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